6. The discrete random variable X has probability generating function

$$G_X(t) = k\ln\left(\frac{2}{2-t}\right)$$

where k is a constant.

(a) Find the exact value of k

(1)

(b) Find the exact value of $\mathrm{Var}(X)$

(7)

(c) Find $\mathrm{P}(X = 3)$

(4)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

6. A discrete random variable X has probability generating function given by

$$G_X(t) = \frac{1}{64}\,(a + bt^2)^2$$

where a and b are positive constants.

(a) Write down the value of $P(X = 3)$

(1)

Given that $P(X = 4) = \frac{25}{64}$

(b) (i) find $P(X = 2)$

(7)

(ii) find $E(X)$

(3)

The random variable $Y = 3X + 2$

(c) Find the probability generating function of Y

(2)

6. The probability generating function of the random variable X is

$$G_X(t) = k(1 + 2t)^5$$

where k is a constant.

(a) Show that $k = \dfrac{1}{243}$ **(2)**

(b) Find $P(X = 2)$ **(2)**

(c) Find the probability generating function of $W = 2X + 3$ **(2)**

The probability generating function of the random variable Y is

$$G_Y(t) = \frac{t(1 + 2t)^2}{9}$$

Given that X and Y are independent,

(d) find the probability generating function of $U = X + Y$ in its simplest form. **(2)**

(e) Use calculus to find the value of $\text{Var}(U)$ **(6)**

6. The discrete random variable V has probability distribution

v	2	3	4
$P(V = v)$	$\frac{9}{25}$	$\frac{12}{25}$	$\frac{4}{25}$

(a) Show that the probability generating function of V is

$$G_V(t) = t^2\left(\frac{2}{5}t + \frac{3}{5}\right)^2$$

(2)

The discrete random variable W has probability generating function

$$G_W(t) = t\left(\frac{2}{5}t + \frac{3}{5}\right)^5$$

(b) Use calculus to find

(i) $E(W)$

(2)

(ii) $\mathrm{Var}(W)$

(4)

Given that V and W are independent,

(c) find the probability generating function of $X = V + W$ in its simplest form.

(2)

The discrete random variable $Y = 2X + 3$

(d) Find the probability generating function of Y

(2)

(e) Find $P(Y = 15)$

(2)

6. The discrete random variable X has probability generating function

$$G_X(t) = \frac{t^2}{(3-2t)^2}$$

(a) Specify the distribution of X

(2)

A fair die is rolled repeatedly.

(b) Describe an outcome that could be modelled by the random variable X

(1)

(c) Use calculus and $G_X(t)$ to find

(i) $E(X)$

(ii) $Var(X)$

(7)

The discrete random variable Y has probability generating function

$$G_Y(t) = \frac{t^{10}}{(3-2t^3)^2}$$

(d) Find the exact value of $P(Y = 19)$

(3)

6. The random variable X has probability generating function $G_X(t)$ where

$$G_X(t) = \frac{1}{\sqrt{4-3t}}$$

(a) Use calculus to find $\text{Var}(X)$
Show your working clearly.
(6)

(b) Find the exact value of $P(X \leqslant 2)$
(4)

The independent random variables X_1 and X_2 each have the same distribution as X

The random variable $Y = X_1 + X_2 + 1$

(c) By finding the probability generating function of Y, state the name of the distribution of Y
(4)

(d) Hence, or otherwise, find $P(X_1 + X_2 > 5)$
(2)

6. The probability generating function of the discrete random variable X is given by

$$G_X(t) = k(3 + t + 2t^2)^2$$

(a) Show that $k = \frac{1}{36}$
(2)

(b) Find $P(X = 3)$
(2)

(c) Show that $\text{Var}(X) = \frac{29}{18}$
(8)

(d) Find the probability generating function of $2X + 1$
(2)